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Experiment-2 : Data Ingestion and Cleaning

Aim:- To acquire data and prepare it for analysis using data ingestion and cleaning techniques

Theory :- Data ingestion refers to the process of importing data from different sources (like CSV files, databases, etc.) into a system for analysis. Cleaning the data is an important step because raw data often contains missing values, duplicates, and inconsistencies.

Key Steps in Data Cleaning :-

Removing duplicate data

Handling missing values

Correcting inconsistent formats

Changing data types

Filtering required data

Tools Used :-

Python

Pandas Library

Programs :-

Basic Level (Weak Students)

1. Load CSV and Display First 5 Rows

```
from google.colab import files
uploaded = files.upload()
import pandas as pd
df = pd.read_csv("/content/data.csv")
import os
print(os.getcwd())

df = pd.read_csv("data.csv")
print(df.head())
```

Choose Files data.csv

data.csv(text/csv) - 7943 bytes, last modified: 5/1/2026 - 100% done

Saving data.csv to data.csv

/content

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

2. Identify Missing Values

```
print(df.isnull().sum())
```

```
total_bill    0
tip           0
sex           0
smoker        0
day           0
time          0
size          0
dtype: int64
```

3. Replace Missing Values with Zero

```
df.fillna(0, inplace=True)
print(df)
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
..	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

[244 rows x 7 columns]

4. Rename Columns to Lowercase

```
df.columns = df.columns.str.lower()
print(df.columns)
```

```
Index(['total_bill', 'tip', 'sex', 'smoker', 'day', 'time', 'size'], dtype='object')
```

5. Display Dataset Shape

```
print("Rows and Columns:", df.shape)
```

```
Rows and Columns: (244, 7)
```

Moderate Level (Normal Students)

6. Remove Duplicate Rows

```
df = df.drop_duplicates()
print(df)
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4
..	...	...	...	...	...	...	...
239	29.03	5.92	Male	No	Sat	Dinner	3
240	27.18	2.00	Female	Yes	Sat	Dinner	2
241	22.67	2.00	Male	Yes	Sat	Dinner	2
242	17.82	1.75	Male	No	Sat	Dinner	2
243	18.78	3.00	Female	No	Thur	Dinner	2

```
[243 rows x 7 columns]
```

7. Handle Missing Values Using Mean

```
df.fillna(df.mean(numeric_only=True), inplace=True)
print(df.head())
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

8. Change Data Types

```
df['total_bill'] = df['total_bill'].astype(int)
```

9. Filter Rows Based on Condition

```
filtered = df[df['total_bill'] > 18]
print(filtered)
```

	total_bill	tip	sex	smoker	day	time	size
2	21	3.50	Male	No	Sun	Dinner	3
3	23	3.31	Male	No	Sun	Dinner	2
4	24	3.61	Female	No	Sun	Dinner	4
5	25	4.71	Male	No	Sun	Dinner	4
7	26	3.12	Male	No	Sun	Dinner	4
..	...	...	...	...	...	...	...
237	32	1.17	Male	Yes	Sat	Dinner	2
238	35	4.67	Female	No	Sat	Dinner	3
239	29	5.92	Male	No	Sat	Dinner	3
240	27	2.00	Female	Yes	Sat	Dinner	2
241	22	2.00	Male	Yes	Sat	Dinner	2

[104 rows x 7 columns]

10. Save Cleaned Dataset

```
df.to_csv("cleaned_data.csv", index=False)
```

Advanced Level (Bright Students)

11. Compare Missing Value Strategies

```
mean_fill = df.fillna(df.mean(numeric_only=True))
median_fill = df.fillna(df.median(numeric_only=True))
print("Mean Strategy:\n", mean_fill.head())
print("Median Strategy:\n", median_fill.head())
```

Mean Strategy:

	total_bill	tip	sex	smoker	day	time	size
0	16	1.01	Female	No	Sun	Dinner	2
1	10	1.66	Male	No	Sun	Dinner	3
2	21	3.50	Male	No	Sun	Dinner	3
3	23	3.31	Male	No	Sun	Dinner	2
4	24	3.61	Female	No	Sun	Dinner	4

Median Strategy:

	total_bill	tip	sex	smoker	day	time	size
0	16	1.01	Female	No	Sun	Dinner	2
1	10	1.66	Male	No	Sun	Dinner	3
2	21	3.50	Male	No	Sun	Dinner	3
3	23	3.31	Male	No	Sun	Dinner	2
4	24	3.61	Female	No	Sun	Dinner	4

12. Clean Inconsistent Categorical Values

```
df['sex'] = df['sex'].str.lower().replace({
    'male': 'M', 'm': 'M',
    'female': 'F', 'f': 'F'
})
```

13. Automate Cleaning Using Function

```
def clean_data(df):
    # Make an explicit copy to avoid SettingWithCopyWarning
    df = df.copy()
    df = df.drop_duplicates()
    df.fillna(df.mean(numeric_only=True), inplace=True)
    df.columns = df.columns.str.lower()
    return df
df = clean_data(df)
```

14. Outlier Detection (IQR Method)

```
Q1 = df['total_bill'].quantile(0.25)
Q3 = df['total_bill'].quantile(0.75)
IQR = Q3 - Q1
outliers = df[(df['total_bill'] < Q1 - 1.5*IQR) | (df['total_bill'] > Q3 + 1.5*IQR)]
print(outliers)
```

	total_bill	tip	sex	smoker	day	time	size
59	48	6.73	M	No	Sat	Dinner	4
102	44	2.50	F	Yes	Sat	Dinner	3
142	41	5.00	M	No	Thur	Lunch	5
156	48	5.00	M	No	Sun	Dinner	6
170	50	10.00	M	Yes	Sat	Dinner	3
182	45	3.50	M	Yes	Sun	Dinner	3
197	43	5.00	F	Yes	Thur	Lunch	4
212	48	9.00	M	No	Sat	Dinner	4

15. Data Cleaning Report

```
print("Data Cleaning Report")
print("Shape:", df.shape)
print("Missing Values:\n", df.isnull().sum())
print("Duplicates:", df.duplicated().sum())
```

```
Data Cleaning Report
Shape: (237, 7)
Missing Values:
total_bill    0
```

```
tip          0
sex          0
smoker       0
day          0
time        0
size         0
dtype: int64
Duplicates: 0
```

Conclusion

In this experiment, we learned how to ingest data from CSV files and perform essential data cleaning operations such as handling missing values, removing duplicates, and correcting inconsistencies. These steps improve data quality and make it suitable for accurate analysis and decision-making.

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